From Dempwolff–Müller to the MCM Permutation Polynomial

1 The MCM Polynomial

Let $F = \mathbb{F}_{2^n}$ and define the truncated trace

$$L_m(x) = \sum_{i=0}^{m-1} x^{2^i}.$$

The Müller–Cohen–Matthews polynomial formalized in the project is

$$\mathrm{MCMpol}_{n,m}(x) = L_m(x)^{2^m+1} x^{(2^n-1)-2^m}.$$

For $x \neq 0$, this is the quotient

$$\mathrm{MCMpol}_{n,m}(x) = \frac{L_m(x)^{2^m+1}}{x^{2^m}},$$

because $x^{2^n-1} = 1$. The polynomial form gives the intended value at zero.

Theorem 1 (MCM permutation theorem)**.** *Assume* $0 < m < n$, m *and* n *are odd, and* $\gcd(m,n) = 1$. *Then* $\mathrm{MCMpol}_{n,m}$ *is a permutation of* F.

This is the Lean theorem `Dobbertin1999.MCM.mcm_permutation`.

2 Two Permutation Inputs

The proof combines a Dempwolff–Müller permutation with a Gold permutation.

2.1 Dempwolff–Müller transfer

For $1 < m < n$, m odd, and $\gcd(m,n) = 1$, let k' satisfy

$$(2^{n-1} - 2^{m-1} - 1)k' \equiv 2^{m-1} \pmod{2^n - 1}. \tag{DM}$$

Then

$$D_{k'}(x) = L_m(x)x^{k'}$$

is a permutation of F. In Lean this is `dempwolff_mueller_permutation_ktransfer`, which reuses `DempwolffMueller.LxXk'_bijective`.

2.2 Gold

Under the oddness and coprimality hypotheses, the Gold map

$$G(y) = y^{2^m+1}$$

is a permutation of F. This is `gold_permutation`, reusing `KasamiAPN.gold_pow_bijective`.

3 The Linking Exponent

The key arithmetic lemma `KasamiAPN.exists_linking_exp` supplies an exponent k' satisfying both (DM) and

$$k'(2^m+1) \equiv (2^n-1) - 2^m \pmod{2^n-1}. \tag{MCM}$$

The second congruence gives, for every $x \in F$,

$$x^{(2^n-1)-2^m} = x^{k'(2^m+1)}.$$

It follows that

$$\begin{aligned} \mathrm{MCMpol}_{n,m}(x) &= L_m(x)^{2^m+1} x^{k'(2^m+1)} \\ &= \left(L_m(x) x^{k'}\right)^{2^m+1} \\ &= G(D_{k'}(x)). \end{aligned}$$

The equality, including the zero case, is formalized as `MCM_eq_dempwolff_mueller_pow`.

4 Proof When $m > 1$

Choose k' from the linking-exponent lemma. By (DM), $D_{k'}$ is bijective. The Gold theorem makes G bijective. Since

$$\mathrm{MCMpol}_{n,m} = G \circ D_{k'},$$

the MCM polynomial is bijective as a composition of bijections. The final Lean step uses `Function.Bijective.comp` after rewriting the function by the factorization theorem.

5 The Case $m = 1$

The transfer theorem requires $1 < m$, but the MCM polynomial is already the identity for $m = 1$. Indeed, $L_1(x) = x$, and

$$\begin{aligned} \mathrm{MCMpol}_{n,1}(x) &= x^{2^1+1} x^{(2^n-1)-2} \\ &= x^{2^n} \\ &= x. \end{aligned}$$

The last equality is the finite-field Frobenius identity. The Lean proof uses `FiniteField.pow_card` and concludes with `Function.bijective_id`.

6 Formal Dependency Summary

`mcm_permutation`	$\longleftarrow$	`MCMpol`, `MCM_eq_dempwolff_mueller_pow`, `KasamiAPN.exists_linking_exp`, `dempwolff_mueller_permutation_ktransfer`, `gold_perm`
`dempwolff_mueller_permutation_ktransfer`	$\longleftarrow$	`DempwolffMueller.LxXk'_bijective`;
`gold_permutation`	$\longleftarrow$	`KasamiAPN.gold_pow_bijective`;
`MCM_eq_dempwolff_mueller_pow`	$\longleftarrow$	`FiniteField.pow_card_sub_one_eq_one`.